



BANNARI AMMAN INSTITUTE OF TECHNOLOGY

(An Autonomous Institution Affiliated to Anna University, Chennai)

SATHYAMANGALAM - 638 401

Module Test - I - Apr - 2026

II Semester

22GE002 & COMPUTATIONAL PROBLEM SOLVING

1	(i)	A retail chain observes frequent stock-outs in their stores despite regular inventory audits. Store managers report delayed deliveries, inaccurate stock records, and lack of demand forecasting. Break down the stock-out issue using scenario decomposition, identify two sub-problems, and explain how solving them individually improves efficiency. (3 Marks - [E/P, 2])
2	(ii)	A school is organizing its annual sports day with 500 students participating. Teachers face challenges in scheduling events, assigning referees, allocating equipment, and arranging different sports venues. Students and parents often get confused about timings, locations, and participation criteria. Decompose this problem into three sub-problems that can be addressed individually. (2 Marks - [An/M, 1])
3	(i)	A program is designed to find the sum of digits of a number. However, an incorrect operator is used. <pre> BEGIN n ← 123 sum ← 0 WHILE n > 0 DO digit ← n / 10 sum ← sum + digit n ← n / 10 END WHILE PRINT sum END </pre> Identify the incorrect operator(s) and replace them with the correct ones. (3 Marks - [An/M, 2])

4	(ii)	A program is designed to print numbers starting from 1 using a loop. The loop continues as long as the value of the variable is less than or equal to 5. <pre> BEGIN i ← 1 WHILE i ≤ 5 DO PRINT i i ← i + 1 END WHILE END </pre> Identify the loop termination condition. When will the loop stop executing? (2 Marks - [Ap/P, 1])
5	(i)	A software development team struggles to deliver a mobile application on time. Issues include unclear client requirements, team miscommunication, and inefficient task allocation. Decompose the project delays into sub-problems, select one sub-problem as priority, and suggest a method to resolve it. (3 Marks - [Ap/P, 2])
6	(ii)	A food delivery company notices that orders are frequently delivered late. Customers complain about delayed deliveries, while delivery personnel report unclear route assignments and long waiting times at restaurants. Using scenario decomposition, identify the main root cause of delays and one sub-problem that should be addressed first. (2 Marks - [An/P, 1])
7	(i)	A school system records the marks obtained by a student in an exam. The system needs to evaluate whether the student has passed or failed based on the marks entered. If the marks are 50 or above, the student passes; otherwise, the student fails. Write a pseudocode for the above scenario (3 Marks - [U/C, 2])
8	(ii)	A pseudocode to multiply a number in a loop is given below. <pre> BEGIN result ← 1 FOR i ← 1 TO 5 DO result ← result × 2 END FOR PRINT result END </pre> Predict the output. (2 Marks - [R/F, 1])

9	(i)	A digital library system stores book IDs in a sorted list for fast searching. When a user searches for a specific book ID, the system does not check each ID one by one. Instead, it repeatedly checks the middle element of the list and decides whether to search in the left half or right half. This process continues until the required book ID is found or the search space becomes empty. Given sorted book IDs: [101, 115, 128, 134, 146, 159, 172, 185, 193, 207] Apply decomposition to solve the problem. Break the process into steps and find the position of the book ID 172 by showing the steps clearly. (5 Marks - [Ap/C, 3])
10	(i)	An algorithm to find the factorial of a given number is given below Step 1: Start the algorithm. Step 2: Read the given number n. Step 3: Set the value of fact to 1. Step 4: Set the value of i to 1. Step 5: Check whether i is less than or equal to n. Step 6: Multiply fact by i and store the result in fact. Step 7: Increase the value of i by 1. Step 8: Repeat Steps 5 to 7 until i becomes greater than n. Step 9: Display the value of fact. Step 10: Stop the algorithm. Draw an equivalent flowchart for the above algorithm. (5 Marks - [U/C, 3])
11	(i)	A program is required to sort a list of 10 numbers in ascending order. The developer finds it difficult to directly write the solution due to multiple steps involved in sorting. To solve this efficiently, the problem needs to be broken down into smaller manageable steps. Apply the steps of decomposition to the above scenario and explain how each step can be used to sort the numbers. (5 Marks - [E/C, 3])
12	(i)	A digital display system generates a special number sequence for a light pattern. The sequence is defined as follows: The first two numbers are 0 and 1 Each next number is the sum of the previous two numbers The system asks the user to enter the number of terms (n) If $n \leq 0$, display "Invalid Input" Otherwise, generate and display the sequence up to n terms Also calculate and display the sum of all generated numbers Draw a flowchart to represent the above system using a loop, clearly showing: Input and validation Initialization of first two terms Loop to generate the sequence Calculation of sum Output of sequence and total sum (5 Marks - [An/P, 3])

13	(i)	A fitness app tracks the daily steps of a user for 10 days: [7500, 10200, 8400, 12000, 9800, 15000, 11000, 7000, 9000, 13000] The app needs to: Calculate the total steps. Find the average steps. Count the number of days the user crossed 10,000 steps. Identify the maximum steps taken in a day. Write an algorithm using a for loop to perform all the above operations. (5 Marks - [An/M, 3])
14	(i)	An online examination system manages student login, verification, monitoring, and result evaluation. The system performs the following steps: The user logs in and undergoes identity verification. If verification fails, the exam session is terminated. If successful, the exam starts and the system monitors for suspicious activity. If suspicious activity is detected, the system checks its severity: Low severity → Warning is given, and the exam continues. High severity → Exam is paused and the administrator is notified. If no suspicious activity occurs, the student continues the exam. When time ends, the exam is automatically submitted and evaluated. If marks are above the passing criteria → Display "Pass" Otherwise → Display "Fail" Draw a flowchart for the above scenario. (5 Marks - [Ap/P, 3])
15	(i)	A mobile app developer wants to display a list of 12 contact names in alphabetical order in the app's contact list. Currently, the names are unsorted, making it difficult for users to locate a contact quickly. Write a step-by-step algorithm using the Bubble Sort technique to sort the contact names alphabetically. Include comparisons, swaps, and iterations until the list is fully sorted. (5 Marks - [E/M, 3])
16	(i)	A delivery company tracks daily parcel deliveries for a week (7 days). For each day, the system asks the user to input the number of parcels delivered and time (in minutes) spent delivering. The system calculates: Total parcels delivered in the week Average delivery time per valid day Highest single-day parcel delivery It also categorizes weekly performance based on total parcels delivered: Less than 200 → Low 200 to 499 → Medium 500 to 799 → High 800 and above → Very High If the user enters negative parcels or time, the program should print "Invalid Input" and skip that day in calculations. Write an algorithm to implement the above weekly tracking and calculations. (5 Marks - [An/M, 3])

17	(i)	3. A hospital management system is designed to assign the nearest available ambulance to an emergency location. The system maintains a list of ambulances and checks their current positions to determine the closest one for faster response time. Given the pseudocode to find the nearest ambulance, identify the key operation that influences the time complexity of the algorithm and justify your answer. <pre> function find_nearest_ambulance(ambulances, emergency_location): nearest_distance = INFINITY nearest_ambulance = null for each ambulance in ambulances: distance = calculate_distance(ambulance.location, emergency_location) if distance < nearest_distance: nearest_distance = distance nearest_ambulance = ambulance return nearest_ambulance </pre> (3 Marks - [U/C, 2])
18	(ii)	1. A university system processes student records where the number of operations increases exactly in proportion to the number of students entered. For example, if 100 students are processed, about 100 operations are performed; if 1000 students are processed, about 1000 operations are performed. Analyze the scenario and determine the time complexity of the algorithm. Justify your answer based on order of growth. (2 Marks - [U/C, 1])
19	(i)	A software company needs to find the maximum sales value from a large list of daily sales for the past week: Daily Sales (in \$): [250, 420, 180, 560, 300, 410, 150] The team decides to use a divide and conquer approach to find the maximum efficiently. Identify the maximum sales value using divide and conquer. Indicate how the original list can be divided into sublists for the first step. (3 Marks - [U/C, 2])
20	(ii)	A search engine needs to quickly locate a specific page number from a sorted list of page IDs containing millions of entries. To improve efficiency, the system repeatedly divides the list into halves and searches only the relevant part. Identify the algorithmic approach used and indicate why it is suitable for this application. (2 Marks - [U/C, 1])
21	(i)	A data processing system uses different algorithms to handle increasing input sizes. Some algorithms execute in constant time, while others show linear, quadratic, or exponential growth in the number of steps as the input size increases. For a given input size of $N = 100$, select any three types of time complexities from $O(1)$, $O(\log N)$, $O(N)$, $O(N^2)$, and $O(2^n)$. Estimate the approximate number of steps involved for each and compare their performance. (3 Marks - [U/C, 2])

22	(ii)	In a data processing application, Algorithm A has time complexity $O(1)$, and Algorithm B has time complexity $O(\log n)$. However, Algorithm A involves a large constant number of operations, while Algorithm B has a very small constant factor. Evaluate whether Algorithm A will always perform better than Algorithm B for all input sizes. Justify your answer using asymptotic analysis and practical performance considerations. (2 Marks - [U/C, 1])
23	(i)	A fitness tracking app records the number of steps walked each day over a period of time: [3000, 4500, 5200, 2800, 6100, 4000, 3500, 4800] To quickly calculate the total number of steps, the app uses a divide and conquer approach by splitting the list into smaller parts, computing partial sums, and combining them. Identify the sum of elements in the left and right sublists after the first division. Identify the final total number of steps. (3 Marks - [U/C, 2])
24	(ii)	A data processing system handles a large list of elements using a divide and conquer strategy. The system repeatedly splits the list into two equal parts at each step until smaller sublists are formed for processing. During execution, the system reaches a stage where there are exactly 4 sublists, each representing a portion of the original data. Indicate the original number of elements in the list. (2 Marks - [U/C, 1])
25	(i)	A company is developing a program to place flags on a square grid for a display competition. The rules are: - No two flags can be in the same row. - No two flags can be in the same column. - No two flags can be on the same diagonal. The program tries all possible arrangements one by one to find a valid placement. Currently, it works on an 8×8 grid, but the company is considering larger grids, such as 10×10. Analyze the program in terms of: - Maximum number of arrangements it may need to check. - Average number of arrangements it will check before finding a solution. - How increasing the grid size affects time and space required. (5 Marks - [U/C, 3])

26	(i)	A cargo ship is loading high-priority containers with a maximum carrying capacity of 12 kg. Each container has a specific weight and profit value. The goal is to maximize the total profit using the Fractional Knapsack approach. <table><tr><td>Items</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>Weight (kg)</td><td>4</td><td>2</td><td>3</td><td>6</td><td>1</td></tr><tr><td>Profits</td><td>20</td><td>12</td><td>18</td><td>24</td><td>10</td></tr></table> Step 1 is done for you. Complete: Step 2: Arrange items in descending order of P_i/W_i Step 3: Select items without exceeding capacity to maximize profit Step 1: Calculate P_i/W_i <table><tr><td>Items</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>P_i/W_i</td><td>5</td><td>6</td><td>6</td><td>4</td><td>10</td></tr></table> (5 Marks - [Ap/C, 3])	Items	1	2	3	4	5	Weight (kg)	4	2	3	6	1	Profits	20	12	18	24	10	Items	1	2	3	4	5	P_i/W_i	5	6	6	4	10
Items	1	2	3	4	5																											
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P_i/W_i	5	6	6	4	10																											
27	(i)	A text processing program searches for a pattern p of length m within a text t of length n using a nested loop approach. Analyze the algorithm to determine its worst-case time complexity, space complexity, and how changing the outer loop to for (int i = 0; i < n; i++) affects its performance. <pre>for (int i = 0; i <= n - m; i++) { int j = 0; while (j < m && t[i+j] == p[j]) { j++; } if (j == m) return i; } System.out.println("No match found"); return -1;</pre> (5 Marks - [U/C, 3])																														
28	(i)	A fuel transportation company has a tanker with a limited capacity of 1000 liters. The company needs to transport different types of fuel, each with a specific volume and value per liter. The goal is to maximize the total value of fuel transported without exceeding the tanker's capacity. This is a classic example of the Fractional Knapsack Problem, where fractions of items can be included. Formulate an algorithm to fill the tanker with the maximum value of fuel without exceeding its capacity by considering the following conditions: a) Fractions of fuel can be included (e.g., 50% of a fuel type). b) The total volume of the selected fuels must not exceed the tanker's capacity. c) The goal is to maximize the total value of the selected fuels. (5 Marks - [U/C, 3])																														

29

(i)

A delivery service needs to deliver parcels to four locations: A, B, C, and D. The distances (in km) between the locations are shown in the graph:

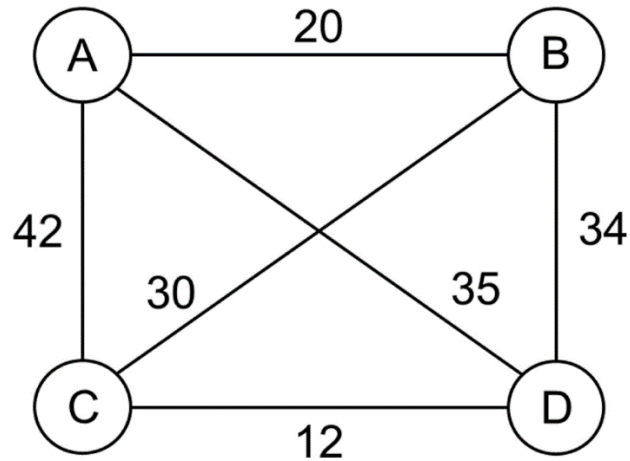
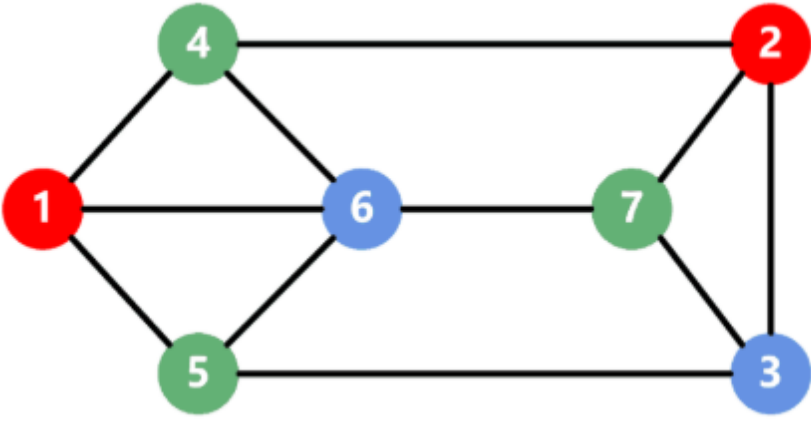


Figure: Graph

The delivery person wants to start from location A, visit each location exactly once, and return to A. Using the brute force approach, determine all possible routes starting and ending at A, calculate the total distance for each route, identify the route(s) with the minimum total distance, and explain why this approach guarantees finding the shortest route.

(5 Marks - [U/P, 3])

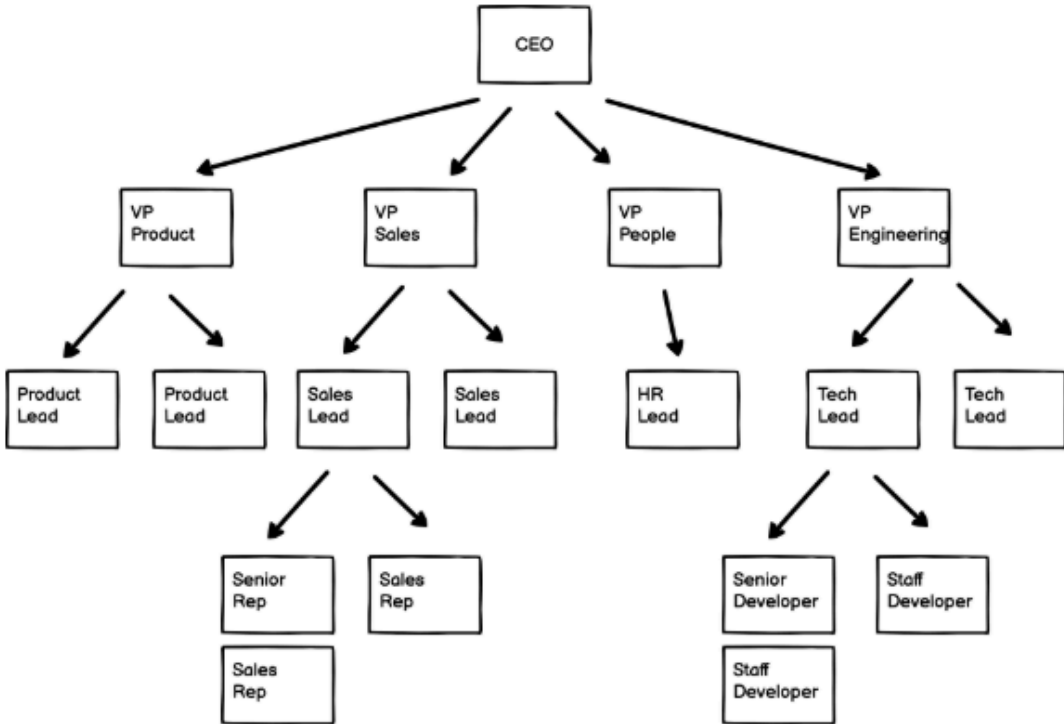
30	(i)	Consider designing a wireless network system where multiple transmitters must be placed in a grid layout. Each transmitter is represented by a queen, and each grid position is represented by a cell on a chessboard. The goal is to place the transmitters such that no two interfere with each other (i.e., no two are in the same row, column, or diagonal). Given the visual representation of a solution for a 4-transmitter setup in the figure titled <i>4-Queens</i>, provide any one valid arrangement for a 5-transmitter system without interference. (5 Marks - [U/C, 3])
31	(i)	A security firm is testing the strength of a high-security briefcase that uses a 3-digit combination lock. Each digit can be any number from 0 to 5. A "Brute Force Attack" involves a specialized machine trying every single possible combination until the lock opens. The machine takes 2 seconds to attempt a single combination. (5 Marks - [U/C, 3])

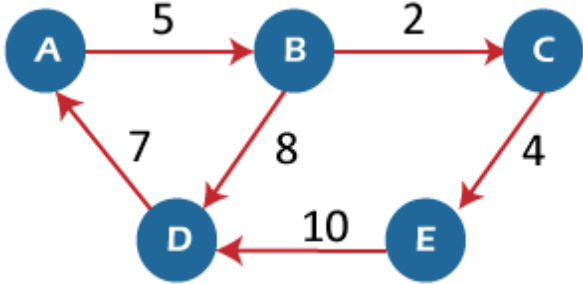
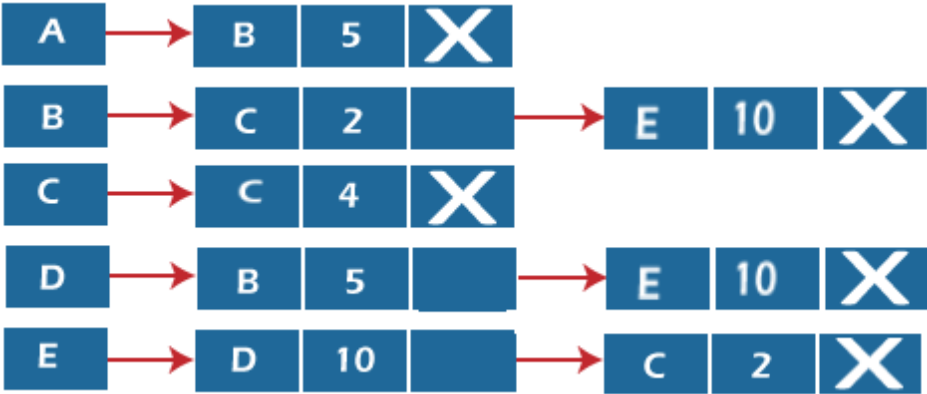
32	(i)	A network administrator is assigning different communication channels to a set of interconnected devices. Each device is represented as a vertex in the graph shown, and an edge between two vertices indicates that the devices are directly connected and cannot share the same channel due to interference. The graph consists of 7 vertices (1 to 7) connected as shown in the figure.  Using a graph colouring technique, assign colours (channels) to all the vertices such that no two adjacent vertices share the same colour. Determine the minimum number of colours required and provide a valid colouring for the graph. (5 Marks - [An/C, 3])
33	(i)	A sales application stores customer information as comma-separated strings. Minor formatting changes cause runtime errors during string splitting and indexing. Developers propose reorganizing the data using structured records to eliminate dependency on string formats. Assess how reorganizing program logic from delimiter-based strings to record-based data reduces errors and improves robustness. (3 Marks - [E/P, 2])
34	(ii)	A social networking platform has millions of users. Each user can have multiple friends, and the platform needs to perform tasks like: - Suggesting new friends based on mutual connections - Quickly identifying friends of friends - Displaying a network of connections for a user - Finding communities or clusters of connected users Each user is connected to other users in a complex network that doesn't follow a simple sequential order. The number of connections varies per user, and relationships are dynamic – new friendships are added and old ones removed regularly. Identify the category of data structure suitable for representing these user relationships and justify. (2 Marks - [U/C, 1])

35	(i)	A bus stop maintains a queue for passengers waiting to board. Initially, the queue contains the following passenger IDs in order of arrival (front → rear): [201, 202, 203, 204] The following actions occur: A new passenger with ID 205 arrives and joins the queue. The first passenger in the queue boards the bus and leaves. Determine the passenger at the front of the queue and list all passenger IDs in the queue after performing the above operations. (3 Marks - [Ap/C, 2])
36	(ii)	A library uses a stack to manage returned books. The most recently returned book is placed on top of the stack. State which stack operation is performed when a librarian adds a newly returned book and briefly describe its effect. (2 Marks - [U/C, 1])
37	(i)	A report-generation algorithm remains unchanged, yet its execution time drops significantly after reorganizing input data into structured records with logical grouping. This improvement surprises junior developers. Assess how data organization alone can enhance algorithm performance without modifying algorithmic steps. (3 Marks - [An/C, 2])
38	(ii)	A weather monitoring system collects daily readings for temperature, humidity, wind speed, and rainfall at multiple locations. Each reading is a single value recorded in the system's memory for further analysis and reporting. Identify the most suitable type of data used to store these individual measurements. (2 Marks - [E/P, 1])
39	(i)	Consider four different real-world situations where tasks or items need to be processed in a sequence. Predict which type of queue would be most suitable and justify briefly. In a fast-food restaurant drive-thru, cars line up to place their orders. The first car to arrive is the first car to be served, and cars move forward as orders are completed. In a library book drop-off system, books can be returned either at the main entrance or at the back entrance, depending on where the librarian is stationed. In a video streaming service, multiple videos are queued to pre-load. When the last video finishes pre-loading, the system cycles back to the first video to continue buffering, ensuring smooth playback. In a fire station emergency call system, calls are attended based on the urgency of the situation rather than the time the call was received. The most critical emergencies are handled first. Example : Simple Queue: Cars are served in the order they arrive, following the FIFO principle. (1 mark) (3 Marks - [An/C, 2])

40	(ii)	A hostel maintains a singly linked list of room numbers to track occupied rooms. When a new student checks in, their room number is added to the end of the list. State which linked list operation is performed when a new room number is added at the end and briefly describe what it does. (2 Marks - [An/P, 1])
41	(i)	Consider an Abstract Data Type (ADT) representing a function to compute the square of a non-negative integer. Implementation in C: <pre>#include <stdio.h> int square(int n) { // Precondition: n >= 0 return n * n; // Postcondition: Returns the square of n } int main() { printf("%d\n", square(-4)); printf("%d\n", square(0)); printf("%d\n", square(6)); return 0; }</pre> Identify which function calls in main() satisfy the precondition and which violate it. Explain the postcondition for the function square and what it guarantees if the precondition is satisfied. (5 Marks - [U/P, 3])

42	(i)	Imagine a company's organizational chart, where the CEO is at the root of the hierarchy. The CEO oversees multiple Department Heads (Level 1). Each Department Head manages several Managers (Level 2), and each Manager supervises a team of Employees (Level 3). The structure helps visualize the reporting relationships within the company. Consider the following tree representing an organizational chart with dummy alphabet representation: <pre> graph TD A((A)) --> B((B)) A --> C((C)) B --> D((D)) B --> E((E)) B --> F((F)) C --> G((G)) C --> H((H)) E --> I((I)) E --> J((J)) G --> K((K)) </pre> Infer the following from the given Tree: 1. Child Node of "D" 2. Sibling of node "E" & "F" 3. Leaf nodes 4. Nodes at Level 2 5. Number of Edges in the tree (5 Marks - [An/P, 3])
43	(i)	Consider an Abstract Data Type (ADT) representing a digital music playlist, where each song has a title, artist, and duration. The ADT supports operations such as creating a playlist, adding new songs, deleting songs, updating song details, and checking information about the songs. Tasks: i) Identify and classify the operations described above into create, copy, destroy, modification, and inquiry operations. Explain the importance of inquiry operations in this ADT and why they do not change the playlist. (5 Marks - [An/C, 3])

44	(i)	Consider a company's organizational hierarchy represented as a tree, where the CEO is at the root, and each subsequent level of employees is arranged as branches. The root (CEO) has direct reports (like department heads), and each department head manages a set of employees.  <pre> graph TD CEO[CEO] --> VP_Product[VP Product] CEO --> VP_Sales[VP Sales] CEO --> VP_People[VP People] CEO --> VP_Engineering[VP Engineering] VP_Product --> Product_Lead1[Product Lead] VP_Product --> Product_Lead2[Product Lead] VP_Sales --> Sales_Lead1[Sales Lead] VP_Sales --> Sales_Lead2[Sales Lead] VP_People --> HR_Lead[HR Lead] VP_Engineering --> Tech_Lead1[Tech Lead] VP_Engineering --> Tech_Lead2[Tech Lead] Sales_Lead1 --> Senior_Rep[Senior Rep] Sales_Lead1 --> Sales_Rep[Sales Rep] Tech_Lead1 --> Senior_Developer[Senior Developer] Tech_Lead1 --> Staff_Developer1[Staff Developer] Tech_Lead2 --> Staff_Developer2[Staff Developer] </pre> Construct a similar tree to represent a university's academic structure in a proper hierarchy. (5 Marks - [E/P, 3])
45	(i)	A library system keeps track of borrowed books using a linked list. Each node contains the book ID and student ID. The current list of borrowed books is: [BookID: 201, StudentID: 11] → [BookID: 202, StudentID: 12] → [BookID: 203, StudentID: 13] Operations performed: A new book BookID: 204 borrowed by StudentID: 14 is added at the end. The book with BookID: 202 is returned and removed from the list. Display the list of currently borrowed books. Tasks: i) Draw or write the linked list after performing the operations. (3 marks) ii) State the final sequence of Book IDs in order. (2 marks) (5 Marks - [An/C, 3])

46	(i)	Consider a delivery network where different cities are connected by various transportation routes, such as highways, railways, or flights. Each city is represented as a node, and each direct route between two cities is represented as an edge. The weight of each edge represents the time (in hours) required to travel between two cities. Cities: A - Delhi B - Mumbai C - Bangalore D - Chennai E - Kolkata  Spot the error and correct it in the below given figure titled Adjacency List.  (5 Marks - [An/P, 3])
47	(i)	Consider an application: Music Playlist – Storing the names of songs in a playlist. The choice of array for the above given application can be One-Dimensional array as it stores a sequential list of song names. Representation of One-Dimensional array: songs = ["Song A", "Song B", "Song C", "Song D"] Justify with reasoning the choice of array type and its representation that is required for the following application scenarios: Application 1: Cinema Seat Booking System – Storing seat availability in a cinema hall. Application 2: Weather Simulation Model – Storing temperature, humidity, and wind speed at different altitudes, locations, and times. (5 Marks - [U/P, 3])

(i)

Consider a flight network where airports are connected by direct flights. Each airport is represented as a node, and each direct flight between two airports is represented as an edge. The network can be represented as a graph where the adjacency matrix provides a convenient way to store and analyze the connections between airports.

A- Delhi Indira Gandhi Airport

B- Mumbai Chhatrapati Shivaji Airport

C- Bangalore Kempegowda Airport

D- Chennai International Airport

E- Kolkata Subhas Chandra Bose International Airport

Figure titled "Graph Representation of Flight Network" represents the Graph of Flight Network of the airports are connected by direct flights.

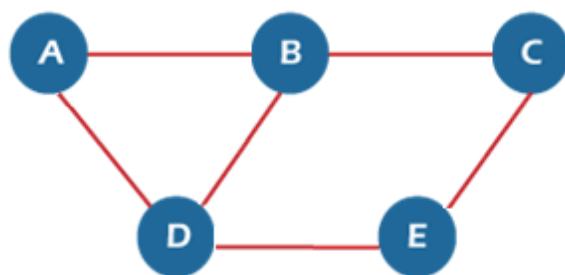


Figure: graph representation

i) Complete the respective adjacency matrix of the flight network given in the below figure titled "Adjacency Matrix of Flight Network".

	A	B	C	D	E
A	0	1	0	1	0
B			1?		
C	0	1	0	0	1
D	1	1	0	0	1
E			2?		

Figure: Adjacency Matrix of Flight Network

ii) If suppose the adjacency matrix value for C turns as C: 0 1 0 1 0, redraw the corresponding graph representation.

(5 Marks - [Ap/P, 2])